

Adaptive Self-Consistency for Long-Form Factuality on Long-Tail Facts

Aggregation, metric sensitivity, and audited budget routing

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Fluent long answers can hide many factual failures

The setting

- Entity-centric questions require **long, coherent answers**.
- Each answer contains many independently checkable claims.
- Rare entities make parametric recall less reliable.

A convincing answer
many atomic claims



Unsupported claims
remain persuasive

long tail

Fluency $\neq$ factual precision

Core tension

More samples can reveal correct facts. A fixed large budget still wastes compute after gains level off.

Three questions organize the study

1

Does sampling help?

Across head, torso, and tail entities.

2

How should candidates combine?

Synthesis versus voting and selection.

3

Can compute be routed?

Preserve factuality while reducing samples.

Empirical thesis

LLMAgg improves the executed FactOWL score; budget conclusions depend on the metric, and adaptive routing remains exploratory.

Evaluation: LLM-judged FactOWL score across tiers

RiDiC benchmark

- 3 domains: rivers, cars, disasters.
- 2 languages: English and Chinese.
- Wikipedia pageview tiers: **head**, torso, **tail**.
- Public pool: up to 99 completions per input.

Same generator: Llama 3.1 8B Instruct

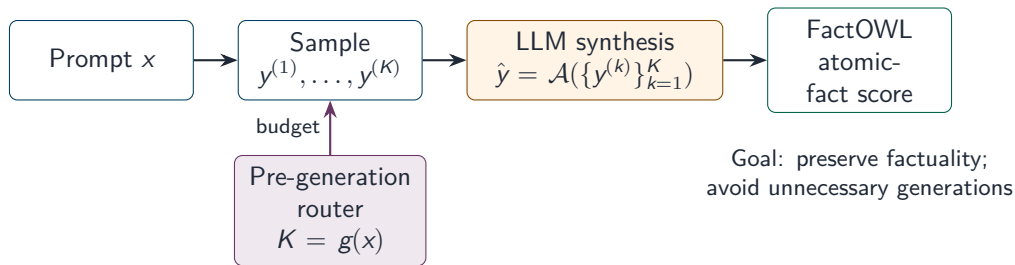
Reported score ($\gamma = 10$)

$$p(y) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[f_i \text{ supported}],$$
$$S_{10}(y) = p(y) \begin{cases} 1, & n > 10, \\ \exp(1 - 10/n), & n \leq 10. \end{cases}$$

Atomic facts and support labels are LLM-judged against the supplied Wikipedia page. For $n = 0$, $S_{10}(y) = 0$.

Lang.	Domain	Tier	RiDiC entity example
EN	cars	tail	Jiotto Caspita
EN	rivers	head	Blue Nile
EN	disasters	torso	Super Outbreak
ZH	cars	tail	劳斯莱斯 Silver Wraith
ZH	disasters	torso	2024 年花莲地震
ZH	rivers	head	伏尔塔瓦河

Method: sample, synthesize, score, then route the budget



- **Single:** candidate 1 from the same fixed pool.
- **LLMAgg:** merge consistent claims and remove contradictions.
- **Adaptive:** exploratory pre-generation choice of K .

Synthesis has two distinct regimes

Complementary evidence

- Correct claims distributed across candidates.
- Agreement helps recover a shared core.
- Synthesis can exceed every input response.

Correlated error

- Candidates repeat related false claims.
- Agreement is not evidence of truth.
- Aggregation can amplify the wrong signal.

Selection inherits one response; LLMAgg constructs a new response from the pool.

Baselines separate agreement, selection, and synthesis

Condition	What it does	Deployable?
Single	First completion from the same fixed pool	yes
MC	Majority co-occurrence over responses	yes
MF	Majority voting over normalized atomic facts	yes
Best-of- K logprob	Same selector on the matched $K = 20$ pool	yes
Best sampled	Oracle picks factuality-best candidate after scoring	no
LLMAgg	LLM synthesizes a new response from the pool	yes

Fair comparison

Single is candidate 1 of the shared pool. Matched- K methods use the same first 20 candidates; full-budget results use all 99.

At matched $K = 20$, synthesis leads selection baselines

	Single	Best-of- K	ReASC	LLMAgg@20
English macro	0.281	0.304	0.290	0.577
Chinese macro	0.324	0.375	0.331	0.564

Across six language-domain cells

- Same first-20 candidate pool.
- LLMAgg leads both selection policies.
- Gains are broad, not driven by one domain.

Interpretation

These are executed S_{10} scores; they are not plain supported-fact precision.

Operational K thresholds depend on the metric

Executed S_{10} : K_{95}/K_{99}

	EN	ZH
Cars	4/8	8/16
Disasters	8/20	8/16
Rivers	8/16	8/16

Raw support $p(y)$: K_{95}/K_{99}

	EN	ZH
Cars	1/1	1/8
Disasters	1/1	2/4
Rivers	1/1	4/16

K_q : earliest tested $K \in \{1, 2, 4, 8, 16, 20\}$ reaching $q\%$ of the $K = 20$ score. No formal K_{sat} is estimated.

Aggregation improves tail entities, but the head-to-tail gap remains

Lang.	Tier	Single	LLMAgg@20
		macro S_{10}	macro S_{10}
EN	head	0.445	0.828
EN	torso	0.384	0.762
EN	tail	0.243	0.513
ZH	head	0.480	0.757
ZH	torso	0.396	0.675
ZH	tail	0.275	0.501

Stable pattern

Head > torso > tail remains visible after improved inference.

Implication

Aggregation raises executed score broadly; long-tail difficulty remains.

Single is candidate 1 from the shared pool; values are macro averages over three domains.

The original $K = 99$ controller is not held-out evidence

Audit findings

- Eight baseline rows created per topic.
- Split performed row-wise, not by topic.
- Nearly every test topic appears in training.
- Raw score joins duplicate repeated titles.

Consequences

- Feature importance is not interpretable.
- Bootstrap rows are not independent.
- Reported $> 99\%$ retention is not a generalization result.
- Savings of 2–29% remain descriptive only.

The adaptive headline is withdrawn pending a topic-grouped rerun.

Deduplicated $K = 20$ reruns save compute but miss targets

Observed range

- Average K : **6.1–8.9**.
- Generation savings: **56–70%**.
- Score retention: **0.85–0.88**.
- Target coverage: **0.40–0.52**.

Why a range?

- Repeated titles have conflicting saved scores.
- Keep-first versus keep-last changes exact cells.
- Zero-score baselines make ratios undefined.
- The negative conclusion is unchanged.

Compute falls, but the requested 0.95/0.99 retention is not achieved.

Adaptive routing remains an open problem

Supported after audit

- Deduplicated reruns reduce average K .
- Some topics benefit substantially beyond $K = 20$.
- Example: Debby, $0.000 \rightarrow 0.622$ from $K = 20 \rightarrow 99$.

Per-topic curves are non-monotonic.

Not supported

- Reliable 0.95/0.99 retention.
- Generalization of the $K = 99$ policy.
- Detection of aggregation failures.
- A universal saturation budget.

Use grouped splits and explicit zero handling.

Three takeaways

- ❶ **Synthesis leads on the executed S_{10} score.**
Single is candidate 1 from the same fixed pool.
- ❷ **Budget thresholds are metric-dependent.**
No formal or universal K_{sat} is established.
- ❸ **Adaptive routing is not yet validated.**
Deduplicated K20 reruns save compute but miss retention targets.

Answer to the opening tension

Aggregation can help; efficient routing requires leakage-free evaluation.

References

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Thank you

Questions?

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Backup slides

Audited tables, definitions, and case traces

Matched- $K = 20$ comparison by language and domain

Lang.	Domain	Single	LLMAgg@20	Gain
EN	rivers	0.211	0.463	+0.252
EN	cars	0.344	0.655	+0.311
EN	disasters	0.287	0.612	+0.325
ZH	rivers	0.269	0.442	+0.173
ZH	cars	0.349	0.633	+0.285
ZH	disasters	0.354	0.616	+0.262

Reading the table

Values use executed S_{10} on the same matched subset. Single is candidate 1 from the first-20 pool.

Supplementary baselines at matched $K = 20$

Lang.	Domain	Single	Best-of- K logprob	ReASC	LLMAgg@20
EN	cars	0.344	0.366	0.354	0.655
EN	disasters	0.287	0.324	0.305	0.612
EN	rivers	0.211	0.221	0.210	0.463
ZH	cars	0.349	0.407	0.349	0.633
ZH	disasters	0.354	0.417	0.368	0.616
ZH	rivers	0.269	0.302	0.277	0.442

Audit note

ReASC is the corrected post-hoc baseline: completion-level FactOWL scores replace the earlier topic-level join.

Best-of- K logprob and ReASC are compared with LLMAgg at the same $K = 20$ budget.

Required protocol for a valid adaptive rerun

Well-defined quantities

$$\text{compute savings} = 1 - \frac{\sum_i K_i^{\text{adaptive}}}{N K_{\text{full}}}$$

$$\text{aggregate retention} = \frac{\sum_i S_{10}(\hat{y}_i^{\text{adaptive}})}{\sum_i S_{10}(\hat{y}_i^{\text{full}})}$$

$$\text{coverage} = \Pr[S_i^{\text{adaptive}} \geq \tau S_i^{\text{full}} \mid S_i^{\text{full}} > 0]$$

Leakage controls

- Group split by language, domain, topic.
- One row per (topic, K) score.
- Report zero-score baselines separately.
- Bootstrap unique topics, not fold rows.
- Freeze metric before model selection.

Generation-count savings still exclude aggregation-token cost; report both when deployment cost matters.

Dataset filtering and FactOWL details

Saved evaluation rows

	rivers	cars	disasters
English	1000	997	1000
Chinese	723	400	723

Counts are saved evaluated rows, not a fresh recount of cached nonempty contexts.

Executed FactOWL score

$$F(y) = \{f_1, \dots, f_n\}$$

$$p(y) = \frac{1}{n} \sum_i 1[f_i \text{ supported}]$$

$$S_{10}(y) = p(y) \begin{cases} 1, & n > 10 \\ e^{1-10/n}, & n \leq 10. \end{cases}$$

Verification uses the supplied page; fact extraction and support labels are LLM-generated proxies. For $n = 0$, $S_{10}(y) = 0$.

Audited case: Debby collapses at $K = 20$, then recovers

Executed S_{10} curve

K	score
1	0.221
2	0.428
4	0.500
8	0.118
16	0.083
20	0.000
32	0.125
64	0.286
99	0.622

What the audit established

- Single is candidate 1 in the pool.
- Published 0.548 pooled facts from all 99 candidates.
- Correct Single score: 0.221; raw support: 3/5.
- No first-20 candidate identifies 2024.
- Wrong years differ, but errors are correlated.

Evidence of correlated error in this setup, not a causal limitation of Llama.